Does the length of an outburst predict the wait for the next? A preregistered forecast test for SS Cygni

Figure and build review preview. Scientific prose is not yet drafted. Barry W. Brook*[1], Claude Opus 5.5 (Anthropic) and GPT-6 Codex (OpenAI)[2]

Figure/build review preview; scientific prose is not drafted. Release and submission are Barry W. Brook's decision.

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Abstract

1. Introduction

2. Evidence

3. Results

4. Discussion

4.1 What follows

4.2 Which conclusions depend on which assumptions

Table 4. Dependence of each main conclusion on its assumptions.

Conclusion	Rests on	Fails if

4.3 What would sharpen the answer

5. Conclusions

Methods in brief

Roles of AI systems and humans

Data and code

References

Fig. 1.

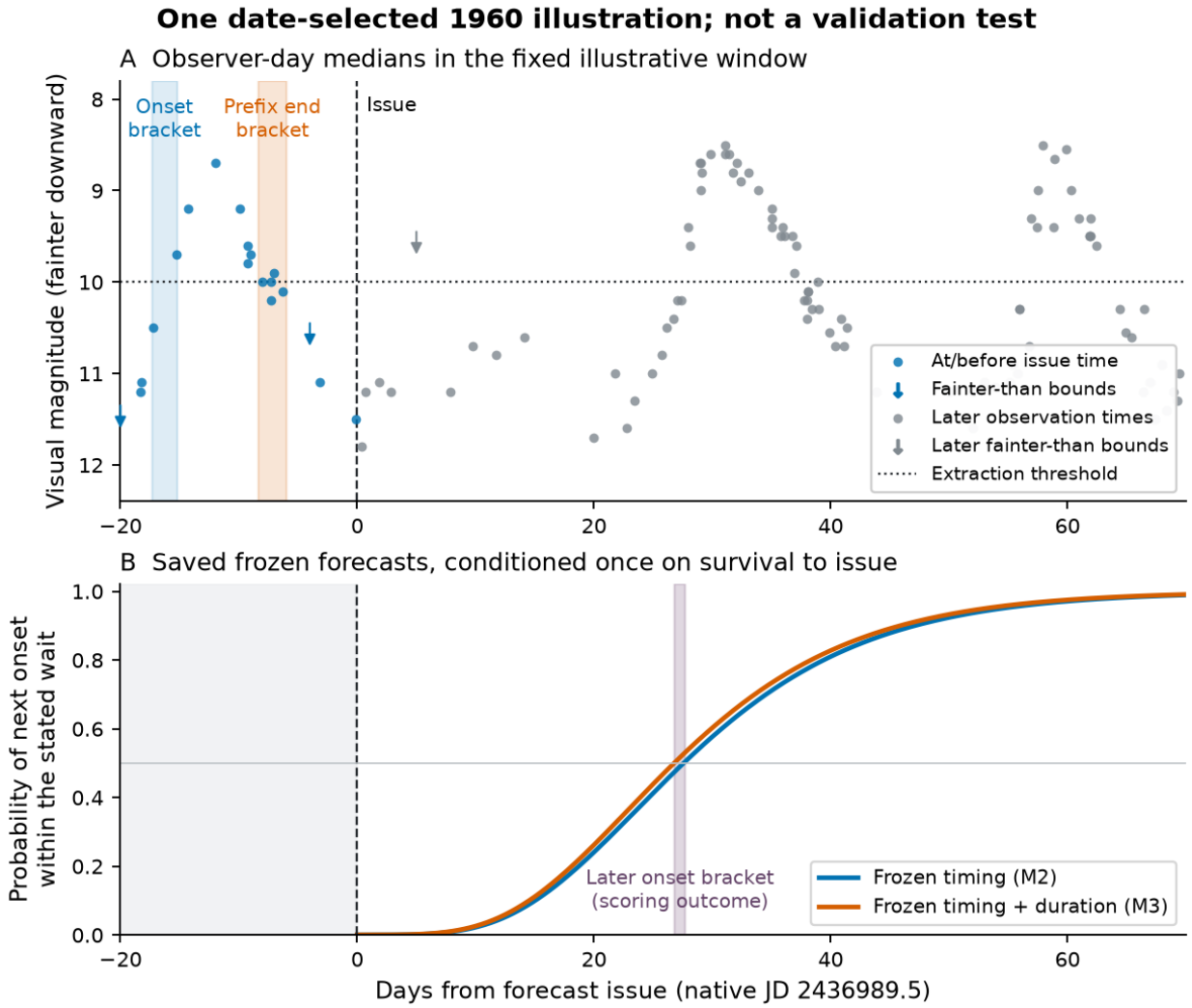


Figure 1: One forecast in the opened SS Cyg record, selected as the first eligible issuance by observation time rather than score. The upper panel shows observer-day display summaries, with detections separated from faint limits; arrows point toward fainter magnitudes. Prefix observations and the fitted crossing brackets define the completed-outburst duration available at issue. Display summaries are calculated separately on each side of issue. The lower panel shows the saved frozen timing and timing-plus-duration distributions, conditioned once on survival to issue. The later onset bracket supplies the scoring outcome. Reported observation times are known; historical upload and revision times are not. This example illustrates the retrospective information set, rather than validating the extraction or forecasting model. The completed comparison followed a post-access format repair.

Fig. 2.

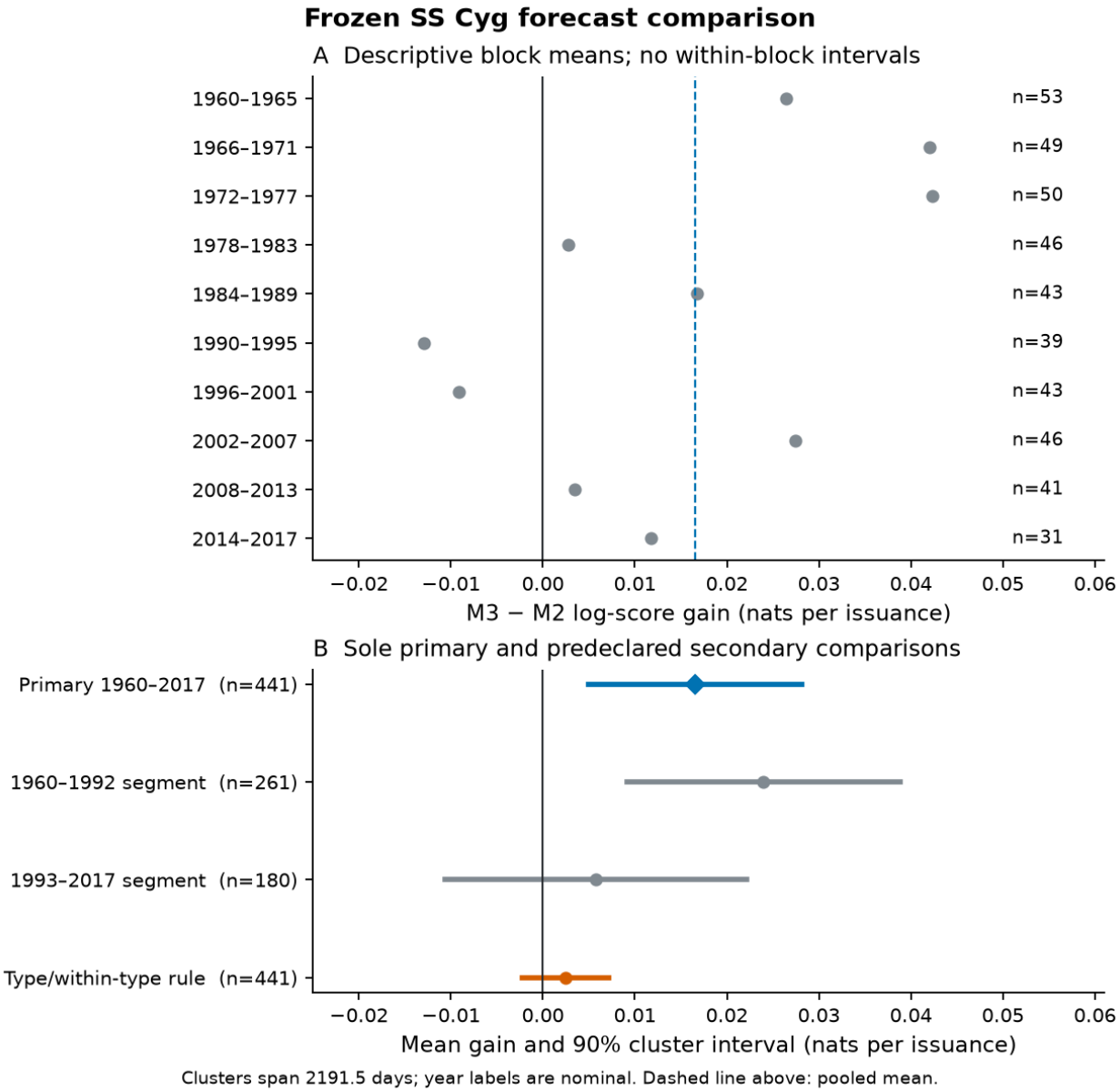


Figure 2: Frozen forecast gain and its predeclared qualifications. The primary paired log-score gain is 0.0165 nats per issuance, with interval [0.0049, 0.0281], over 441 forecasts. Horizontal intervals in the lower panel use the predeclared cluster procedure; only the primary comparison is confirmatory. The later segment and frozen type/within-type rule remain inconclusive. The upper panel shows descriptive means for the fixed blocks, without block-specific confidence intervals; calendar labels are approximate. The result concerns incremental information beyond the specified timing baseline, following the recorded post-access format repair, and does not identify a physical mechanism.

26 **Fig. 3.**

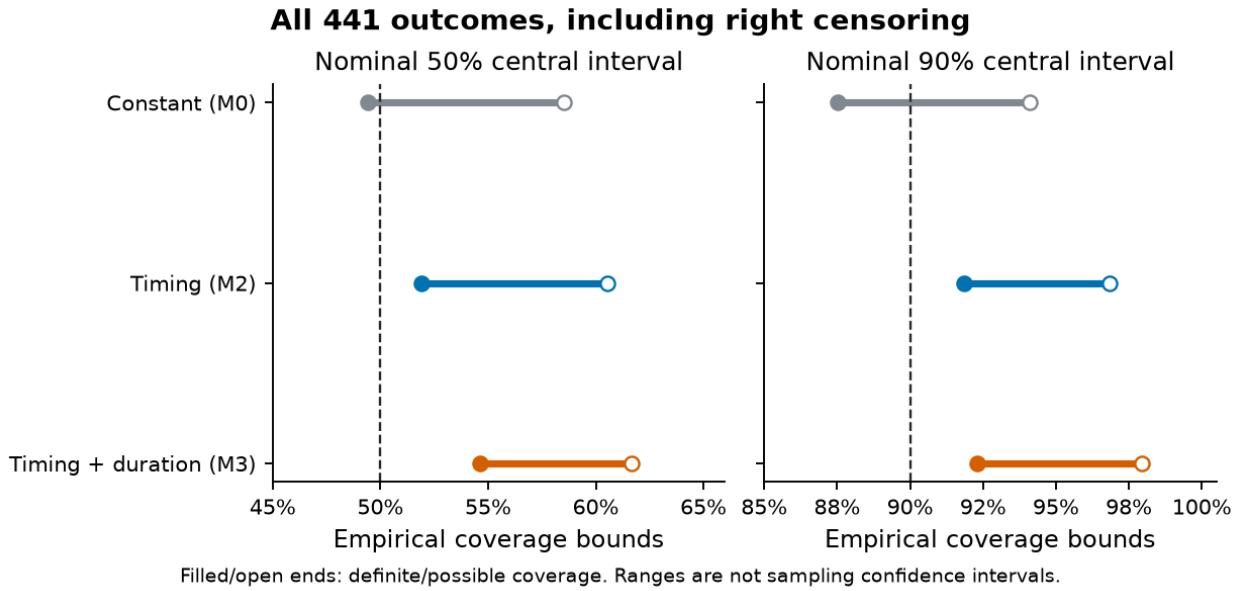


Figure 3: Observed coverage of central predictive intervals for the constant, timing and timing-plus-duration forecasts. Each segment runs from definite to possible coverage given the extracted onset bracket or right censoring; filled and open endpoints distinguish these bounds. Dashed lines mark the nominal fractions. These are partial-identification bounds in the realised sample, conditional on the extraction and scoring population. They are not confidence intervals for population coverage or guarantees about physical onset times. The sample shows overcoverage; the primary paired gain does not imply sharply calibrated operational predictions.

27 **Fig. 4.**

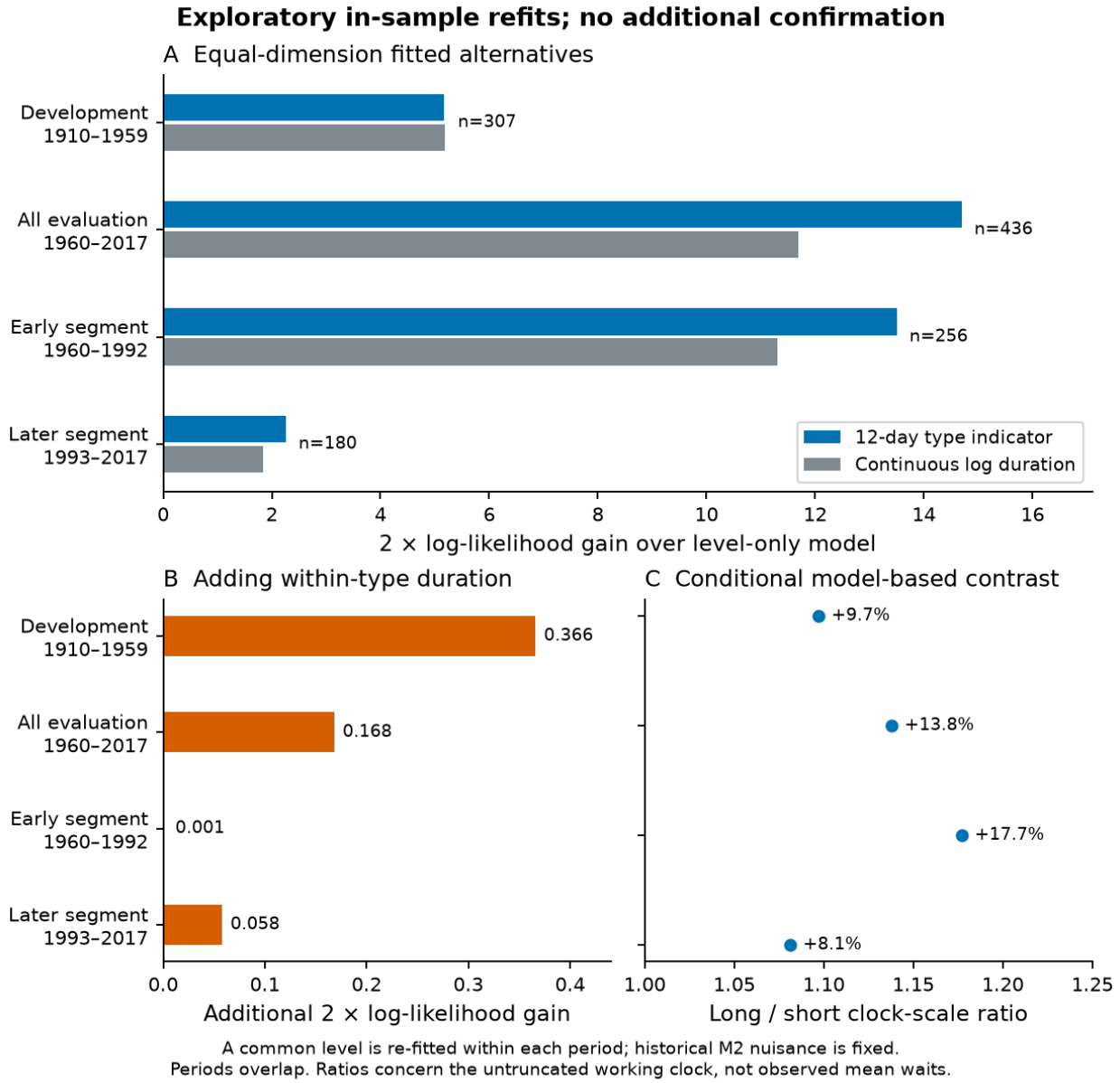


Figure 4: Exploratory post-result comparisons, fitted using the evaluation outcomes. With a free level and historical timing nuisance parameters held fixed, the class-only fit has likelihood-ratio statistic 14.7, compared with 11.7 for continuous log duration; these equal-dimensional models are nonnested. Adding a linear within-class term to class gives 0.17. This is neither exclusive attribution to class nor an absence or equivalence test for within-class effects. The right panel gives conditional long/short ratios of the untruncated working-clock scale, rather than observed-gap or remaining-wait ratios. Era points are descriptive refits, not an era-difference test. No panel supplies a further held-out confirmation.